

Jon Vogel

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Machine Learning Software Engineer

Business minded Principal Engineer & Technical Innovator with a track record of taking nascent ideas, proving viability and rapidly transforming them into robust, high-performance production systems to achieve product market fit or solve a business problem. Former airline pilot and industrial engineer who builds self-governing and fail-safe processes. Delights in contributing to or leading a team with a winning and supportive culture. Smart humble optimist who celebrates ideas and effort across a team through meritocracy. Ever curious and forever learning from others or hands on implementation of cutting-edge tech. Maintains an entrepreneurial mindset to solve complex, cross-departmental problems. Obsessed with software fundamentals such as test-driven development (TDD), extendibility, delivery (CI/CD), security, scalability and maintainability.

PROFESSIONAL EXPERIENCE

NOMAD GO

Redmond, WA

Founding engineer that led the technical execution behind funding events, product pivots, paid POC's, engagements with blue-chip companies (Starbucks, Pepsi Co, Walmart) recurring revenue and deployments to 10,000+ locations. Company grew from 5 to 50 people during tenure. Lead inventor on US Patent 12,002,008 B2 for edge computer vision and Augmented Reality Inventory solution.

Principal Software Engineer (AI)

December 2024 – April 2026

- Improved accuracy by 30% on visual Q & A tasks by fine tuning a multimodal VLM on data curated from iOS application
- Eliminated manual product onboarding by building an AI Native workflow on iOS with tools and structured output
- Reduced reliance on object detection models by building a visual RAG pipeline matching objects at 60fps on the edge
- Enabled best in class reusable AI features by maintaining a Swift Package to interact with the OpenAI and Anthropic API's
- Saved token costs and enabled offline inference by using models from Hugging Face and the Foundation Models SDK
- Evaluated prompt accuracy using unit tests, ground truth data and tolerant assertions

Software Engineer

August 2018 – November 2024

- Architected backend from scratch on AWS using CloudFormation, ECS Fargate, Aurora PostgreSQL, SQS, Cognito, and S3
- Developed Authorization and fine-grained access control of data using JWT's, principals of least access and the shared security model. Solution could scale to 1 million + concurrent queries for less the \$5,000 a month
- Deployed and maintained Hasura's GraphQL solution to remove non-differentiating API work. Wrote RESTful API's and Headless Microservices in Node.js and TypeScript to provide differentiating functionality to client software
- Designed and built native iOS applications on top of our APIs to deliver computer vision solutions to multiple verticals from retail, quick serve, and building control automation
- Collaborated with Machine Learning Engineer to develop CoreML computer vision models to run on "edge" iOS devices
- Developed user less authentication method based upon Apples Device Check and App Attest API's
- Enrolled the company in the Apple business manager program and configured an MDM server to track hardware and provide 100% uptime

UNEARTH TECHNOLOGIES

Seattle, WA

March 2017 – August 2018

Mobile Developer: Delivered key iOS Features to drive customer traction, develop product market fit and help secure \$4 million in funding. Used Swift, Realm, Google maps, WebSockets, and iOS best practices to develop, test and deploy features at a rapid pace. Gained valuable project management and design exposure while contributing to company culture. Sole developer.

- Developed offline mode to allow construction workers seamless low latency experiences
- Flew drones over construction sites to enable volumetric displacement measurements powering core app feature
- Collaborated with front end web developer to ship web-based features in iOS app using novel token exchange mechanism
- Created custom camera experience using AVFoundation and matching UI/UX design specification
- Programmed unique radial menu perfectly matching tricky design

PEBBLEBEE

Seattle, WA

May 2015 – March 2017

iOS Developer: Founding mobile engineer who took ownership from design to deployment to deliver v1 product which was funded by Kickstarter.

- Sole developer for app that achieved 20,000+ users and 4.5 stars on the App Store
- Worked with firmware engineer and used iOS knowledge to drive BLE specifications to "wake up" device in background
- Shared local MAC build box running Jenkins to build CI/CD pipelines alongside Android Engineer

- Delivered pixel perfect UI/UX matching contract designers' specifications

OWNER / FOUNDER

Flicker - <https://flicker.productwheels.com/> - *Evaluate business ideas using standard frameworks and automations*

- Shipped cross platform MacOS / iOS app leveraging on device AI to automate business development
- Used in app subscriptions to sell context driven best in class image generation
- Wrote novel customer discovery features integrated with Reddit and YouTube

Aviation Resources - <https://aviationresources.io/> - *Flight training resource management.*

- App has 400+ monthly and 100+ daily active users
- Managed contract developers through Upwork

Quick Recap - <https://quickrecap.productwheels.com/> - *Professional-grade transcription made simple.*

- Hired and managed development, design and marketing through Upwork
- 1,000 users and 90 days' worth of audio / video summarized

EDUCATION

Bachelor of Science – Business Finance, University of Washington
BIS141 & 142 Programming Data Structures, Cascadia Community College
Certificate iOS Development, Code Fellows

ADDITIONAL EXPERIENCE

Pilot – Horizon Airlines	April 2014 – January 2015
Certificated Flight Instructor – Boeing Employees Flying Association	June 2014 – January 2016
Methods Process Analyst – Boeing Commercial Airplanes	June 2013 – April 2014
Weight and Balance Tech – Boeing Commercial Airplanes	April 2012 – June 2013

KEY SOFTWARE SKILLS

iOS: RealityKit | ARKit | Vision | SwiftUI | CoreML | SwiftData | AVFoundation | Combine | Foundation Models | Accelerate | Core Image | Core Graphics | CreateML | Metal | PDFKit | Swift Concurrency

Cloud/Backend: PostgreSQL on RDS | CloudFormation | EC2 | ECS | IAM | SQS | S3 | Node/Express | PostgreSQL | Docker | Hasura | GraphQL | RESTful API Design | JavaScript | TypeScript | Security and Obfuscation

Tools: GitHub & Git | Pair Programming | Code Reviews | Agile and Scrum | Point Bidding | Ceremonies